

Lesson 8: Composite Functions

Criteria	Exceeds Expectations	Meets Expectations	Below Expectations
Understanding Notation	Consistently interprets $f(g(x))$ and $(f \circ g)(x)$ correctly; understands the "inside-out" process.	Mostly interprets notation correctly, with minor errors in identifying inner/outer functions.	Struggles to identify the order of function application.
Evaluation (Numerical)	Accurately evaluates $f(g(a))$ for any given value a , showing all steps.	Evaluates $f(g(a))$ with minor arithmetic errors.	Fails to apply inputs in the correct order (e.g., g then f).
Algebraic Composition	Correctly substitutes functions $f(g(x))$, simplifies completely, and maintains accuracy.	Substitutes functions correctly, but makes minor algebraic errors (e.g., distribution, squaring).	Unable to correctly substitute one function into another.
Domain Analysis	Identifies the domain of $f(g(x))$, including restrictions from both functions.	Identifies most of the domain, missing only complex restrictions.	Fails to consider domain restrictions of the inner function.
Modeling & Problem Solving	Expertly applies composite functions to real-world scenarios, explaining the result.	Uses composite functions to solve problems with some guidance.	Unable to connect the concept to application problems.